A VISUAL GUIDE

Pain-Face

Measuring pain from a face and a pulse,
using nothing but an ordinary camera

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Companion to the Pain-Face Technical Report

The problem worth solving

Pain is measured by asking. “On a scale of nought to ten, how bad is it?” That works well, right up until the patient cannot answer.

A sedated patient in intensive care cannot answer. Neither can someone with advanced dementia, a newborn, a patient recovering from anaesthesia, or someone whose stroke has taken their speech. For these people, pain is judged by a nurse watching the face, intermittently, between other duties. It is skilled work, and it is subjective, and it stops the moment the nurse leaves the room.

The obvious idea is to let a camera watch instead. The difficulty is that a camera watching a face is easily fooled.

Why the face alone is not enough

The face carries genuine pain information. Decades of research established which muscles move during pain, and a standard score, **PSPI**, built from them. But that score is not pain-**specific**:

- A person wincing at a bad smell activates several of the same muscles.
- So does someone straining to lift something heavy.
- So, awkwardly, does a broad smile: two of the muscles that score pain points are also the muscles of a genuine grin.
- And a stoic patient in real pain may show almost nothing at all.

Why the pulse alone is worse

The body reacts to pain too. The heart speeds up, beat-to-beat variability falls, breathing quickens, the skin’s small vessels tighten. All of this is measurable from the same camera, with no sensor touching the patient.

But none of it is specific to pain. A racing heart means someone climbed stairs, drank coffee, is anxious about the result, or is simply warm. On its own it is nearly useless as a pain signal.

> **The idea behind Pain-Face**
>
> Neither channel is trustworthy alone, but they fail in **different** ways. The face says **what** is happening: this is the specific pattern of a pain expression, not a smile. The body says **what it is costing**: this person’s physiology is reacting, not just their face.
>
> Put together, each one covers the other’s blind spot.

What you see on screen

Two numbers, side by side



Figure 1: The two readings the system produces. On the left, the dial and the **fused index**, everything combined, on the familiar 0-10 pain scale. Beside it the **PSPI**, the published facial pain score on its own 0-16 scale. The five blocks below step from None to Worst as the reading climbs.

We deliberately show both rather than one. The PSPI is the established, published measure that clinicians and researchers already know, so it stays visible and unmodified. The fused index is our contribution: the facial evidence after correction, adjusted by what the body is doing.

A reader who trusts only the literature can read the PSPI and ignore everything else. Nothing is hidden behind the combined number.

The face, read muscle by muscle

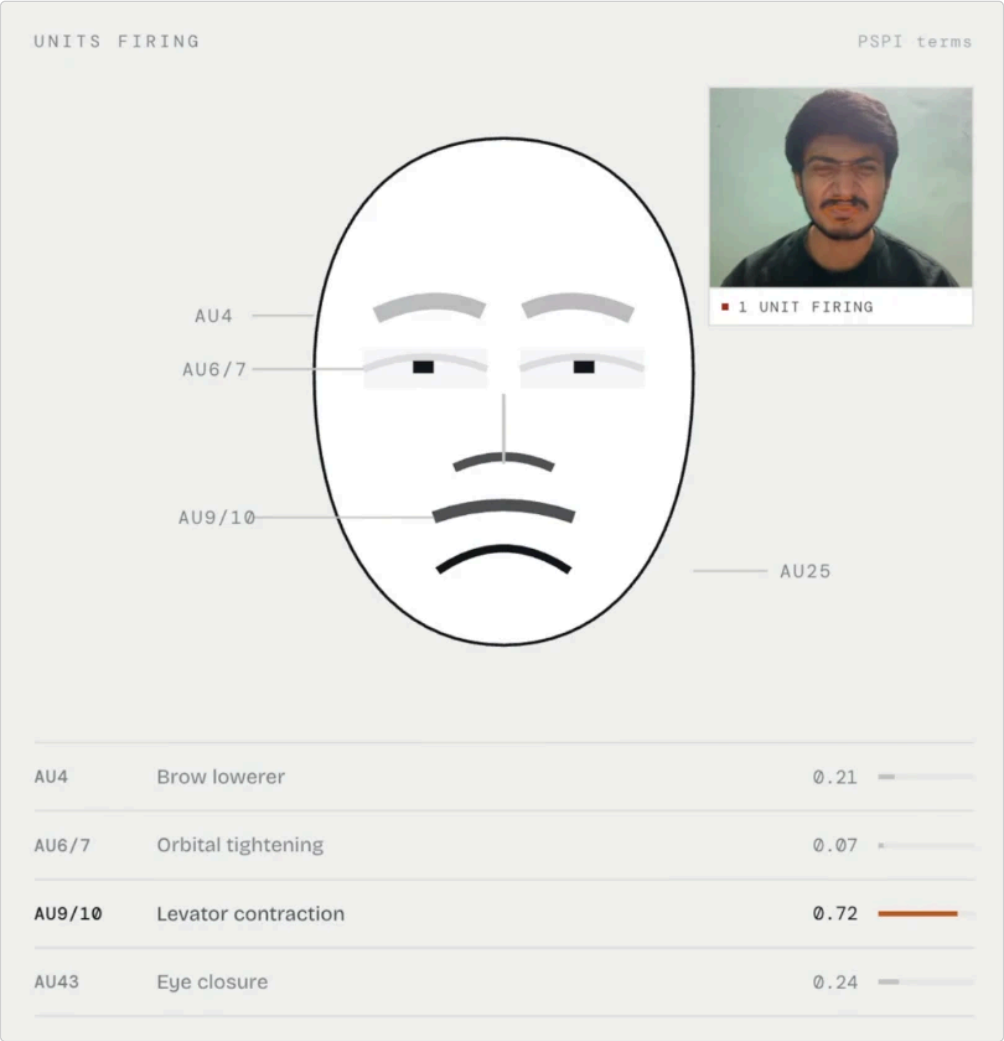


Figure 2: The four components of the PSPI score, drawn on a face diagram rather than listed as numbers. The live camera inset sits in the corner, so the diagram can be checked against the real face at a glance. Here the levator contraction (the nose-and-upper-lip group) is firing strongly at 0.72, while the brow and eye components are quiet.

This panel exists because of a criticism of our earlier work. A system that simply announces “pain detected” cannot be checked. This one shows **which** muscle group produced the score, so a clinician can look at the patient and judge for themselves whether the machine is reading the face correctly.

The four rows are exactly the four terms of the PSPI formula: brow lowering, orbital tightening, levator contraction, eye closure. Nothing else contributes to the score.

Reading a pulse from a face

This is the part that surprises people: the camera measures the heartbeat without anything touching the patient.

Every time the heart beats it pushes blood into the vessels under the skin. That changes, very slightly, how much light the skin absorbs, far too little for the eye to notice, but enough for a camera. By watching patches of skin across many frames and tracking the colour shift, the pulse can be recovered. This is called **remote photoplethysmography**, or rPPG.

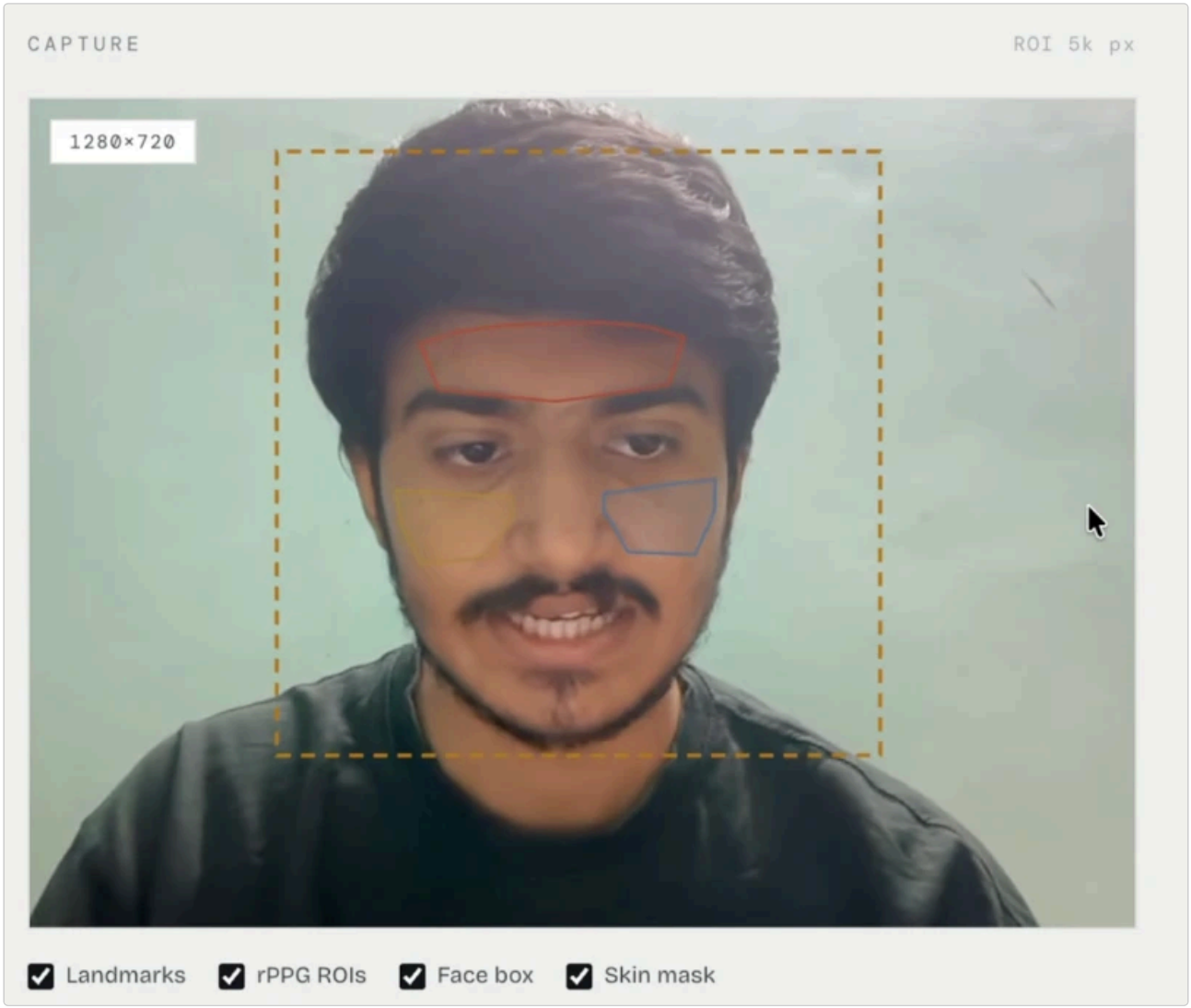


Figure 3: Three patches of skin are tracked continuously: the forehead and both cheeks, chosen because they are well supplied with blood and relatively free of hair and shadow. The dashed box is the detected face. The patches follow the face as it moves.

Only skin-coloured pixels inside those outlines are used; hair, eyebrows and background are filtered out, because they carry no pulse and would dilute the signal.



Figure 4: The recovered pulse waveform and its frequency spectrum. The peak in the spectrum is the heart rate, here 72 beats per minute. This came from an ordinary laptop webcam, with no contact sensor of any kind.

What the body is doing



Figure 5: The autonomic panel. Respiration rate and perfusion are available quickly; the heart-rate-variability measures stay greyed out until enough signal has been collected, rather than showing an unreliable early estimate.

Why there is a “baseline” button

A heart rate of 95 means nothing by itself. For a calm adult it is raised; for someone who just walked upstairs it is ordinary; for an anxious patient it may be their normal.

So the system does not use absolute values. It asks the operator to record a short calm period first, then measures every later reading as a **deviation from that person’s own rest**. A rise of 20 beats above this patient’s baseline is information. A heart rate of 95 is not.

Until a baseline exists, the autonomic panel contributes nothing to the score and says so.

Why combining them is better than either alone

This is the central claim of the project, so it is worth setting out plainly.

The rule: the face must speak first

The combination is deliberately **not** an average of the two channels. Averaging would mean a calm face plus a racing heart produces a moderate pain score, which would be wrong, and dangerous.

Instead, the facial evidence sets the reading, and the body is only allowed to **adjust** it:

> **No pain on the face → no pain reported,**
>
> **however fast the heart is beating.**

We tested this directly. Shown a completely neutral face together with a heart rate of 130 and suppressed variability (physiologically, a person under considerable stress), the system reports **0.2 out of 10**. It refuses to call that pain, because the face does not support it.

When the face **does** show pain, the body then adjusts the reading by up to about a third in either direction:

Situation	What the system does
Pain face, body also reacting	Reading is raised. Face and physiology agree, the strongest evidence available.
Pain face, body completely calm	Reading is lowered. The expression is there, but nothing else is. Worth less confidence.
Neutral face, body racing	Near zero. Something is happening to this person, but there is no evidence it is pain.
No baseline recorded yet	Facial score only, labelled “facial only” on screen so nobody mistakes it for the full reading.

The problem this solves: pain-like faces

Our earlier published study ended by warning that a facial detector “can be misled by ‘pain-like’ expressions”. Here is that problem, caught by the system itself:

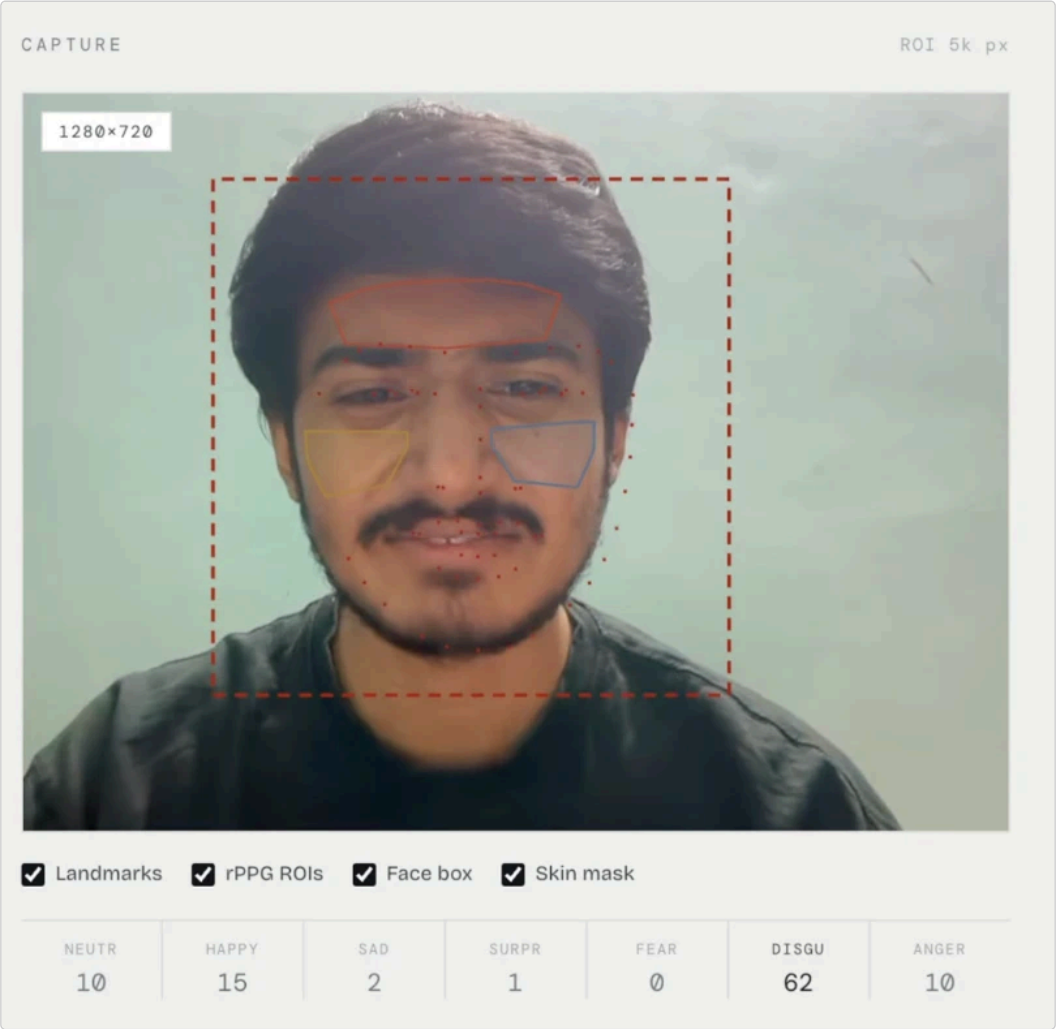


Figure 6: A deliberately disgusted expression, not pain. The emotion row at the bottom reads **disgust 62**. Several of the muscles that score PSPI points are active, because disgust and pain genuinely share them.

The facial channel alone cannot fully resolve this, and we say so in the technical report: disgust and anger remain the system's biggest weakness. What the physiological channel offers is a second, independent opinion: a disgusted face is not usually accompanied by the sustained autonomic response of real pain.

That is the hypothesis the combination is built on. It is a reasonable one, and it is the reason this platform exists, but it has not yet been proven on patients, and the technical report is explicit about that.

The smile case, which we have measured

One pain-like expression we can already quantify is the smile. Two of the muscles that earn PSPI points are also the muscles of a genuine grin, so a smiling person scores high on the raw published formula.

Tested on 95 photographs from a published research database, scored against ratings by human observers:

Faces	Raw PSPI score	After our correction
Happy	69%	**43%**
In pain	89%	**84%**

Smiling faces drop by 26 points; genuinely painful faces barely move. The correction targets the confusion rather than just lowering everything.

Catching pain as it happens

A single frame is a snapshot. Pain is an event, with a beginning and an end.

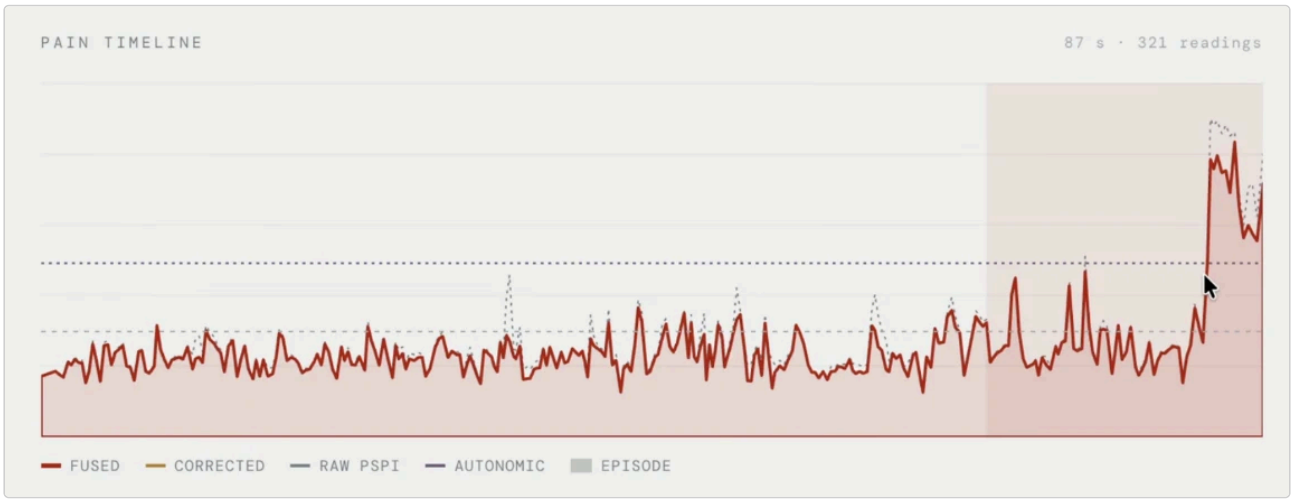


Figure 7: The timeline. The shaded band marks a detected pain episode: the moment the reading rose above threshold and stayed there.

The system does not raise an episode the instant the number crosses a line. A grimace lasting a fraction of a second is not an episode, and a blink should not end one. So the reading must stay high for a short **dwell** period before an episode opens, and must stay low for a longer period before it closes.

EPISODES 1

#	ONSET	DUR	PEAK	MEAN	PEAK AUS
1	69.5s	live	8.7	4.1	**AU04 AU06 AU07 AU09 AU10 AU25**

Figure 8: Each episode is logged with when it began, how long it lasted, its peak and mean, and, most usefully for a clinician, exactly which facial muscles were firing at its worst moment.

That last column is what makes the log worth keeping. “Episode at 69.5 seconds, peak 8.7, driven by AU4, AU6, AU7, AU9, AU10 and AU25” is a reviewable clinical observation. “Pain detected” is not.

When the system refuses to answer

A measuring instrument that quietly reports a wrong number is more dangerous than one that reports nothing. Pain-Face is built to abstain.



Figure 9: The camera was delivering frames too slowly for a pulse to be measured honestly. Rather than printing a plausible-looking heart rate, the system explains why it is withholding one.

Figure 10: The same condition in the status bar. The heart rate simply disappears until the frame rate recovers.

This matters more than it sounds. A pulse cannot be measured reliably below a certain frame rate. Below it, the maths produces a confident number that is simply wrong. We found this the hard way: an early version happily reported 52 beats per minute from a stream far too slow to support it. The system now measures its own frame rate and refuses below the limit.

It abstains in two other places as well: heart-rate variability is withheld until there is enough recording to compute it properly, and “no face in frame” is reported as **no measurement** rather than silently scored as “no pain”.

Not one algorithm, but eight

There is no single agreed-best way to extract a pulse from skin colour. Published methods each make different assumptions, and each fails differently under motion, lighting shifts and skin tone.

METHOD BANK — 8 algorithms

METHOD	HR	SNR
GREEN	58.3	-0.9
ICA	53.4	2.0
PCA	58.9	-3.9
CHROM	76.5	-2.3
PBV	53.7	-2.4
POS	76.8	-2.8
LGI	58.9	-3.8

Figure 11: All eight algorithms run simultaneously on the same signal, each reporting its own heart rate and its own signal quality.

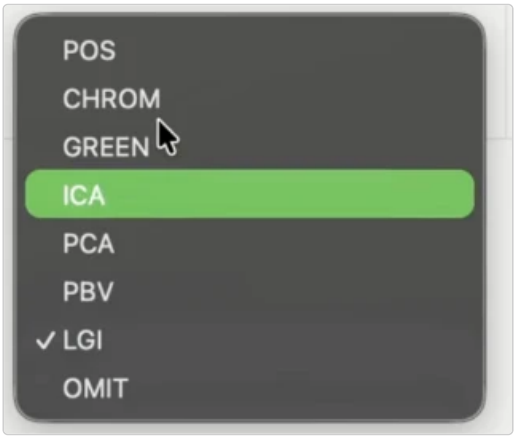


Figure 12: Any method can be selected as the displayed waveform.

Running them together serves two purposes. The final heart rate is a quality-weighted combination rather than one algorithm's opinion, which makes it more robust. And when the eight disagree strongly, that disagreement is itself a warning that the signal is poor.

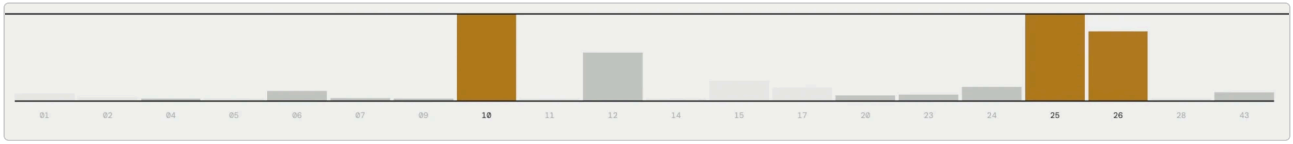


Figure 13: All twenty facial action units the system tracks, shown as a single strip. The pain-relevant units are highlighted when they fire, so the shape of a pain expression becomes recognisable at a glance rather than needing to be read number by number.

Taking the data away with you

A reading on screen is useful in the moment. A study needs the numbers in a file, so everything the system measures can be recorded and exported.

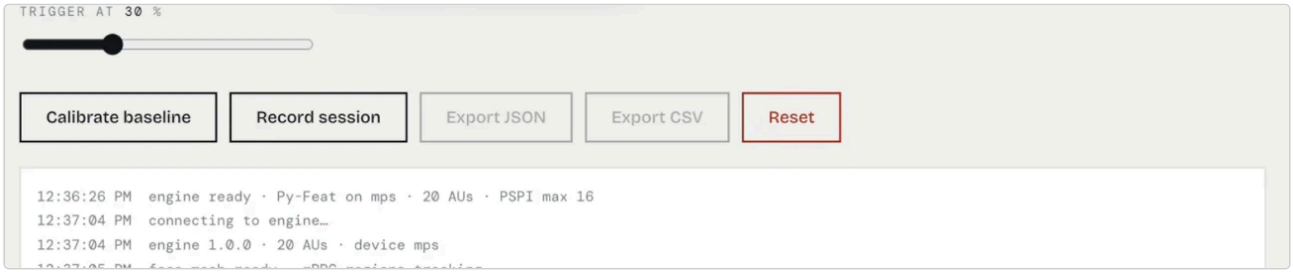


Figure 14: The protocol panel. **Record session** starts keeping every reading; pressing it again stops. **Export JSON** and **Export CSV** then download the recording. The slider sets how strong a reading must be before an episode is declared; the log beneath records what the engine did and when.

Two formats are produced, for two different jobs.

CSV, for statistics software

One row per reading, 46 columns, opening directly in Excel, SPSS, R or pandas. The columns are grouped:

Group	Columns
Timing and scores	`t`, `nrs`, `intensity`, `pspi`, `corrected`, `smile`, `autonomic`, `confidence`, `label`, `triggered`, `quality`
Action units	All twenty: `AU01`, `AU02`, `AU04` … `AU43`
Emotions	`Neutral`, `Happy`, `Sad`, `Surprise`, `Fear`, `Disgust`, `Anger`
Physiology	`hr`, `hr_snr`, `rmssd`, `sdnn`, `lf_hf`, `resp`, `perfusion`, `spo2_ratio`

Because the raw action units and the physiological features are both present, anyone can recompute the pain score with a different formula, or ignore ours entirely and work from the underlying measurements.

JSON, for the whole session at once

The same per-reading samples, plus a summary the system computes for you: how long the recording ran, the peak and mean pain, how much of the session was spent above threshold, how often each pain action unit fired, and the full episode list. An episode entry looks like this:

```
{ "index": 1, "start_t": 2.0, "end_t": 3.5, "duration_s": 1.5,
  "peak_nrs": 7.25, "peak_t": 2.0, "mean_nrs": 3.83,
  "samples": 4, "peak_aus": ["AU04", "AU07", "AU09", "AU10"] }
```

That last field is the one that makes a record reviewable. It says not merely that an episode occurred, but which facial muscles were driving it at its worst moment, which is something a clinician or a coder can check.

One detail worth knowing

Exports are written from the engine's own record, at the full rate it measured, not from what happened to be drawn on screen. Nothing is lost to the display.

Where something like this could be used

Patients who cannot report pain themselves

The clearest case. Sedated intensive-care patients, advanced dementia, neonates and infants, recovery from anaesthesia, and patients whose stroke has affected speech. These groups are consistently under-treated for pain precisely because the standard tool, asking, does not work.

Continuous monitoring on a ward

A nurse checks a post-operative patient every few hours. A camera could watch between those checks and flag episodes for review. Not to replace the nurse's judgement, but to make sure that what happened at three in the morning is not lost by the time of the next round.

Remote and tele-medicine consultations

The consultation already happens over a camera. Nothing extra needs to be worn, attached, or shipped to the patient.

Clinical trials of analgesics

Trials rest heavily on self-reported pain scores, which are subjective and collected only at intervals. A continuous, objective behavioural measure alongside self-report would strengthen that evidence, and because the system logs which muscles fired, the measure is auditable rather than a black box.

Research into pain expression itself

The platform records every action unit, every physiological feature, and the full timeline, exportable as data. That is useful to anyone studying how pain expression varies between people, conditions, or cultures.

> **Where it should not be used**
>
> Anywhere the person being measured has an interest in the result being different. Pain assessment is not lie detection, and this system must not be repurposed to judge whether someone is "really" in pain for the benefit of an insurer, an employer, or a court. It is a clinical aid for people who cannot speak for themselves, not an instrument of suspicion against people who can.

What is proven, and what is not

Being clear about this is the difference between a research tool and a claim.

Claim	Status	Evidence
The system reads facial muscles accurately	**Tested**	Agrees with human expert coders better than the standard comparison tool
The facial score tracks human judgements of pain	**Tested**	Correlation of 0.65 against observer ratings on 95 photographs
The smile correction works	**Tested**	Smiling faces drop 26 points, painful faces 5
The pulse measurement is accurate	**Partly**	Verified against synthetic signals of known rate; never against a real heart monitor
Combining face and body beats face alone	**Not yet**	The central claim, and the next thing to test
It works on real patients	**Not yet**	All testing so far is on photographs and healthy volunteers

The honest summary: the **facial** half of this platform has been measured against human experts and performs well. The **physiological** half works in the engineering sense but has never been checked against a real heart monitor. And the claim that joining them beats either alone, the reason the project exists, is a well-motivated hypothesis that has not yet been tested on people in genuine pain.

Future studies

1. Validate on patient video

Move from photographs to the standard clinical video archive, in which real patients with shoulder injuries were recorded during painful movement and every frame was scored by certified experts. This measures the facial channel against genuine clinical pain rather than posed expressions.

2. Check the pulse against a real sensor

Record volunteers with both the camera and a proper contact heart monitor, at rest and under a controlled, ethically approved painful stimulus. This is what would turn “the pulse feature works” into “the pulse feature is accurate”.

3. Prove the combination earns its place

With the first two done, test directly whether face-plus-body outperforms face alone. If it does not, that is a result worth publishing too.

4. Solve the disgust and anger problem

The current correction handles smiles. Disgust and anger remain confused with pain. The physiological channel is the most promising route, since those expressions should not carry the same sustained bodily response, but it must be shown, not assumed.

5. Test fairly across skin tones

Both facial analysis and camera-based pulse measurement are known to vary with skin tone and lighting. Early results show measurable differences between groups. Any system intended for clinical use must be tested on the full range of patients it will meet, and that testing must be published whatever it shows.

6. Move it to the bedside

Everything here runs on a laptop with a webcam. A ward deployment needs a fixed camera, controlled lighting, a way of handling patients who turn away or are partly covered, and a decision about where the video goes. Processing on the device itself, with only the derived numbers leaving it, avoids sending patient video anywhere and is the approach we would take.